

The Program in Action

When you run *multiplicationtable.py*, the output will look like this:

```
Multiplication Table, by Al Sweigart al@inventwithpython.com
| 0 1 2 3 4 5 6 7 8 9 10 11 12
-----
0| 0 0 0 0 0 0 0 0 0 0 0 0 0
1| 0 1 2 3 4 5 6 7 8 9 10 11 12
2| 0 2 4 6 8 10 12 14 16 18 20 22 24
3| 0 3 6 9 12 15 18 21 24 27 30 33 36
4| 0 4 8 12 16 20 24 28 32 36 40 44 48
5| 0 5 10 15 20 25 30 35 40 45 50 55 60
6| 0 6 12 18 24 30 36 42 48 54 60 66 72
7| 0 7 14 21 28 35 42 49 56 63 70 77 84
8| 0 8 16 24 32 40 48 56 64 72 80 88 96
9| 0 9 18 27 36 45 54 63 72 81 90 99 108
10| 0 10 20 30 40 50 60 70 80 90 100 110 120
11| 0 11 22 33 44 55 66 77 88 99 110 121 132
12| 0 12 24 36 48 60 72 84 96 108 120 132 144
```

How It Works

Line 9 prints the top row of the table. Notice that it sets a large enough distance between the numbers to accommodate products that are a maximum of three digits long. Since this is a 12×12 multiplication table, this spacing can fit the largest product, 144. If you want to create a larger table, you may need to increase the spacing for the columns as well. Keep in mind that the standard terminal window is 80 columns wide and 24 rows tall, so you cannot create much larger multiplication tables without having the rows wrap around the right edge of the window.

```
1. """Multiplication Table, by Al Sweigart al@inventwithpython.com
2. Print a multiplication table.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: tiny, beginner, math"""
5.
6. print('Multiplication Table, by Al Sweigart al@inventwithpython.com')
7.
8. # Print the horizontal number labels:
9. print(' | 0 1 2 3 4 5 6 7 8 9 10 11 12')
10. print('-----')
11.
12. # Display each row of products:
13. for number1 in range(0, 13):
14.
15.     # Print the vertical numbers labels:
16.     print(str(number1).rjust(2), end='')
17.
18.     # Print a separating bar:
19.     print('|', end='')
20.
```